

	Ratio increases.	
6a	$R = \frac{\rho l}{A}$ $= \frac{5.0 \times 10^{-7} \times 2.0}{3.3 \times 10^{-7}}$ $= 3.03 \, \Omega$ $= 3.0 \, \Omega$	M1 A1
6b	$\text{Current through wire} = \frac{1.5}{3.0 + 0.75}$ $= 0.40 \, \text{A}$ $I = nAqv$ $v = \frac{0.40}{8.5 \times 10^{28} \times 3.3 \times 10^{-7} \times 1.6 \times 10^{-19}}$ $= 8.91 \times 10^{-5} \, \text{m s}^{-1}$	C1 A1
6ci	$V_{ST} = \frac{3.0}{3.0 + 0.75} \times 1.5$ $= 1.2 \, \text{V}$ $E = V_{SP} = \frac{1.4}{2.0} \times 1.2$ $= 0.84 \, \text{V}$	M1 A1

6cii	Second wire has higher resistance Potential difference (per unit length) across second wire increases (For the same E), balance length decreases.	M1 M1 A1
7a		